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Comparative bacteriome and antibiotic resistome analysis of water and sediment of the Ganga River of India

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Abstract

A comparative analysis between water and sediment can provide better information to understand the dynamics of the inhabitant microbiome and their respective antibiotic resistance genes of a river. Therefore, the present investigation was carried to explore the limited information available on bacterial microbiome and their predictive antibiotic resistance genes (ARGs) from water and sediment of the Ganga River. The study utilized the NGS-based sequences previously submitted under the accession number (PRJNA847424 and PRJNA892876). Overall analysis revealed that twenty phyla and fifty-four genera were shared between the water and sediment of the Ganga River. Of them, nine phyla and nineteen genera were observed as significantly different (p -value < 0.05). Where the majority of the genera were associated with the sediment samples over the water that identify the sediment samples as more diverse for species richness. Similarly, seventy-six ARGs were shared between water and sediment samples. Of the ten abundant antibiotic resistance pathways, seven were relatively abundant in sediment samples as compared to the water. Vancomycin resistance genes were significantly more abundant among sediment samples, whereas β -lactam resistance genes were equally distributed in water and sediment samples. The network analysis further revealed that five genera (*Flavobacterium*, *Pseudomonas*, *Acinetobacter*, *Candidatus_divison* CL5003, and *Candidatus_division* SWB02) showed a significantly positive correlation with six antibiotic resistance pathways (β -lactam, vancomycin, multidrug resistance, tetracycline, aminoglycoside, and macrolide resistance pathways). The study comes out with several findings where sediment may be considered as a more atrocious habitat for evolving the resistance mechanisms against threatful antibiotics over the water samples of the Ganga River.

Keywords Water and sediment · The Ganga River · NGS · Antibiotic resistance genes (ARGs) · Bacterial-microbiome

Introduction

The Ganga River is a most sacred and holiest river of India with an immense religious, social, spirituals and ritual beliefs. This river is highly exposed to anthropogenic activities as it serves approximately 400 million people with a population growth of ~ 520 person/km² (Singh and Singh 2007; Ali et al. 2021). The river is facing tremendous pollution intake due to increased urbanization, industrialisation, population increase, and agricultural discharge directly or indirectly into the river (Pruden et al. 2012; Bowes et al. 2020). Introduction of the pollutants and toxicants also

affect the dynamics of the microbial communities significantly, where the phenomenon of antibiotic resistance has now emerged as a health threat globally. Several allochthonous load from human sources lead to increased abundance of antibiotic resistance genes (ARGs) (Marti et al. 2013). Approximately 66% of the Ganga River basin are used for agricultural purpose (Saha and Ghosh 2020). Uttar Pradesh region are responsible for approximately 35% (1255.2 MLD) of sewage water discharge into the Ganga River which includes 1081 GPIs (Grossly Polluting Industries), and 913 industries (CPCB 2019–2020). Besides, Uttar Pradesh region has higher agriculture land use and human intervention than the Rishikesh sites of Uttarakhand region (Shukla et al. 2018; Maurya et al. 2021). Therefore, Uttar Pradesh stretch of the Ganga River can be considered as comparatively more exposed to industrial and human interventions than the other sites. The majority of the reports on the microorganisms of the Ganga River are based on cultivation-based

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approach that portray an incomplete image of the bacterial diversity of a habitat (Lata et al. 2009; Niveshika et al. 2016; Dhiman et al. 2018; Matta and Bisht 2018; Chaturvedi et al. 2020). In past few years, a handful investigation has been carried out using metagenomic approaches to explore the bacterial diversity using non-traditional metagenomic approaches. These reports identified the prevalence of bacterial population having bioremediating potential towards major pollutants in the sediments of the Ganga River (Behera et al. 2020; Srivastava and Verma 2023a; Rout et al. 2022). Similarly, few reports also discuss the profile of antibiotic resistance genes in the water and/or/sediment of the Ganga River through metagenomic approaches (Zhang et al. 2019; Reddy and Dubey 2019; Kumar et al. 2021). However, their studies are either seasonal based (Zhang et al. 2019) or limited to one region only (Reddy and Dubey 2019; Kumar et al. 2021) and thus leaves several gaps that need to be addressed. Comparative analysis between water and sediment provides a rationale to understand the microbial dynamics in the interface of water and sediment. To date, only one report is available on comparative analysis of the Ganga River bacteriome and antibiotic resistome between river and sediment. However, it is restricted to one stretch of the Ganga River i.e., Varanasi (Reddy and Dubey 2019) and thus demand for a comprehensive elucidation in this line on a longer region of the river. Therefore, the aim of the present investigation is to compare the profile of bacteriome

and antibiotic resistance genes (ARGs) between water and sediment, where we covered a longer stretch of ~70% of the Ganga River.

Materials and methods

Collection of water and sediment samples of the Ganga River

Illumina generated MiSeq2500 sequences were used from Uttarakhand [Rishikesh (R)] and Uttar Pradesh [Kanpur (K), Allahabad (A), Mirzapur (M), and Varanasi (V)] regions. The place and collection time of water and sediment samples (~50.0 gms; 4–5 cm beneath the river) are mentioned in Srivastava and Verma 2023a; Table 1. The water samples (~5 L) from 20 to 30 cm depth were also collected from the middle of the Ganga River at the same time and place simultaneously i.e., from December 2020 to February 2021 and stored at 4 °C until their processing (Srivastava and Verma 2023b). The Bio-Project accession number PRJNA892876 and PRJNA847424 were used to retrieve the hypervariable 16S V3–V4 sequences from water and sediment samples respectively for comparative analysis at the level of bacterial diversity and antibiotic resistance genes between the samples.

Table 1 Collection of water and sediment samples of the Ganga River

S. No.	Sample type	Latitude	Longitude	States	Region	Sites	Pooled samples	References
1.	Sediment	26.507454N	80.317142E	Uttar Pradesh	Kanpur	Ganga barrage (KS1)	KS	Srivastava and Verma (2023a)
2.	Sediment	26.613820N	80.274830E	Uttar Pradesh	Kanpur	Brahmavart (KS2)		
3.	Sediment	25.425183N	81.886119E	Uttar Pradesh	Allahabad	Triveni Sangam (AS1)	AS	
4.	Sediment	25.502176N	81.855645E	Uttar Pradesh	Allahabad	Rasulabad (AS2)		
5.	Sediment	25.294897N	83.009171E	Uttar Pradesh	Varanasi	Assi (VS1)	VS	
6.	Sediment	25.306138N	83.010457E	Uttar Pradesh	Varanasi	Dashashwamedh (VS2)		Srivastava and Verma (2023b)
7.	Sediment	25.153661N	82.572192E	Uttar Pradesh	Mirzapur	Baria ghat (MS1)	MS	
8.	Sediment	25.152168N	82.563796E	Uttar Pradesh	Mirzapur	Pakka ghat (MS2)		
9.	Sediment	30.122174N	78.314337E	Uttarakhand	Rishikesh	Ram jhula (RS1)	RS	
10.	Sediment	30.102776N	78.300454E	Uttarakhand	Rishikesh	Triveni ghat (RS2)		
11.	Water	26.506911N	80.320077E	Uttar Pradesh	Kanpur	Ganga barrage (KW1)	KW	
12.	Water	26.613131N	80.275771E	Uttar Pradesh	Kanpur	Brahmavart (KW2)		
13.	Water	25.426545N	81.883248E	Uttar Pradesh	Allahabad	Triveni Sangam (AW1)	AW	
14.	Water	25.502685N	81.854575E	Uttar Pradesh	Allahabad	Rasulabad (AW2)		
15.	Water	25.290556N	83.006672E	Uttar Pradesh	Varanasi	Assi (VW1)	VW	
16.	Water	25.307248N	83.012107E	Uttar Pradesh	Varanasi	Dashashwamedh (VW2)		
17.	Water	25.153727N	82.572142E	Uttar Pradesh	Mirzapur	Baria (MW1)	MW	
18.	Water	25.152351N	82.563685E	Uttar Pradesh	Mirzapur	Pakka (MW2)		
19.	Water	30.123621N	78.316734E	Uttarakhand	Rishikesh	Ram jhula (RW1)	RW	
20.	Water	30.102738N	78.304037E	Uttarakhand	Rishikesh	Triveni (RW2)		

Comparative analysis between water and sediment samples

Shared (common) and unique (uncommon) taxa abundance were analyzed to explore and compare the core and unique bacterial population present in water and sediment of the Ganga River samples.

Significantly different phyla and Genera in water and sediment samples and their co-occurrence

Significantly different phyla and genera between water and sediment samples were determined by *T*-test (two-sample) statistical analysis. Further, Spearman correlation was used to determine the co-occurrence pattern of significantly (p -value < 0.05) different phyla and genera obtained in the water and sediment samples of the Ganga River.

Antibiotic resistance pathways and genes analysis

The PICRUSt2 (Phylogenetic Investigation of Communities by Reconstruction of Unobserved States) (version 2.4.1) software was used for predicting functional profiling of the bacterial community prevailing in water and sediment of the Ganga River. KEGG (Kyoto Encyclopaedia of Genes and Genomes) and BRITE (Biomolecular Relations in Information Transmission and Expression) databases were used for profiling of the functional antibiotic resistance pathways and their respective gene prevalence in the water and sediment samples of the Ganga River based on KO orthologs and reads.

PCoA and network analysis

The PCoA (Principal Coordinate analysis) analysis was made by using PAST (Paleontological Statistics Software Package for Education and Data Analysis; 4.03 version) software with the Bray–Curtis dissimilarity index to measure the dissimilarity with respect to bacterial community among water and sediment samples and abundance of antibiotic resistance genes. For network analysis, the Cytoscape (version 3.9.1) software was used (Shannon et al. 2003).

Results

Shared and unique taxon abundance

Average analysis at phyla level revealed that out of 58 assigned phyla, 20 phyla were shared between water and sediment samples. Of which, top 10 phyla include

Proteobacteria (38.3%), Bacteroidota (12.9%), Actinobacteriota (11.1%), Planctomycetota (7.3%), Verrucomicrobiota (5.1%), Acidobacteriota (4.2%), Cyanobacteria (3.3%), Chloroflexi (3.0%), Gemmatimonadota (2.4%), and Patescibacteria (2.1%) (Supplementary File 1). Whereas, the genus level analysis revealed only 54 shared genera out of 882 assigned genera between sediment and water samples. Of them, *Flavobacterium* (6.47%), *Pseudomonas* (3.58%), *Nitrospira* (3.03%), *Candidatus division OM190* (2.02%), *Hydrogenophaga* (1.66%), *Gemmatimonas* (1.66%), *Candidatus division A0839* (1.36%), *Candidatus division OLB12* (1.26%), *Pirellula* (1.25%), *Vicinamibacteraceae* (1.21%) represent the top ten genera (Supplementary File 2). These shared phyla and genera constitute the core bacteriome of the Ganga River water and sediment samples of this investigation (Fig. 1).

Only one phylum (Abditibacteriota) was unique (not shared) in any of the water samples, whereas sediment exhibited twenty unique phyla (Fig. 2; Supplementary File 1). However, the prevalence of all these unique phyla was < 1%.

Similarly, at the genus level, sediment harbours higher count of unique genera i.e., 257 in which *Anaerolinea* (1.03%) is the only genus showing > 1% abundance, rest were present in < 1% abundance (Fig. 2; Supplementary File 2). The water samples consist of only 136 unique genera in which only *Sporichthyaceae* (1.14%) showed > 1% abundance, remaining 135 genera are showing < 1% abundance (Fig. 2; Supplementary File 2).

Significantly different phyla and Genera in water and sediment samples

Of the twenty shared phyla between water and sediment samples, nine were found to be significantly different (p -value < 0.05). It includes *Proteobacteria*, *Actinobacteriota*, *Acidobacteriota*, *Chloroflexi*, *Nitrospirota*, *Desulfobacterota*, *Myxococcota*, *Latescibacterota*, and *Spirochaetota*. Of them, two phyla [(*Proteobacteria* (41.32%) and *Actinobacteriota* (15.52%)] were relatively more prevalent in water samples, whereas the remaining seven phyla were more dominant in sediment samples (Fig. 3A; Supplementary Table S1).

At the genera level, nineteen significantly different genera were observed among 54 shared genera in between water and sediment samples. Of them, only four [*Pseudomonas* (5.54%), *Gemmatimonas* (2.15%), *Candidatus division A0839* (1.77%), and *Candidatus division SMIA02* (1.07%)] showed higher association with the water samples, whereas the remaining fifteen genera were relatively higher in sediment samples (Fig. 3B; Supplementary Table S2).

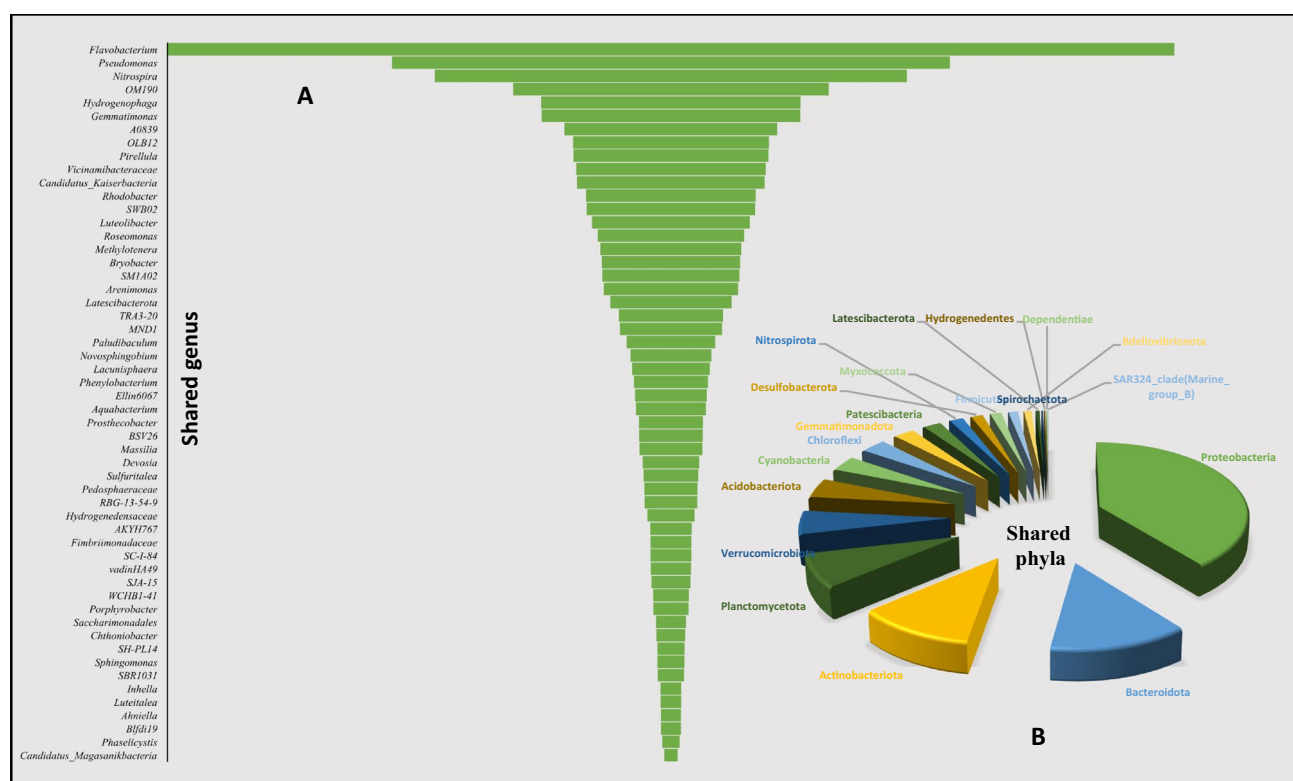


Fig. 1 Abundance of shared genera (A) and phyla (B) between the water and sediment samples of the Ganga River

Co-occurrence of significantly different bacterial taxon

Co-occurrence among significantly different taxa (9 phyla and 19 genera) between water and sediment were studied using Spearman correlation analysis. Proteobacteria showed significant positive correlation with Actinobacteriota (SC: 0.78; p -value: 0.007), whereas it was significantly negative with Chloroflexi (SC: -0.72 ; p -value: 0.018) and Actinobacteriota (SC: -0.80 ; p -value: 0.004), and *Desulfobacterota* (SC: -0.70 ; p -value: 0.021). Nitrospirota showed significant positive correlation with Acidobacteriota (SC: 0.68; p -value: 0.029). *Desulfobacterota* also showed significant negative correlation with Actinobacteriota (SC: -0.76 ; p -value: 0.009), whereas it was positively correlated with Chloroflexi (SC: 0.96; p -value: 0.00). Myxococcota showed positive correlation with Acidobacteriota (SC: 0.63; p -value: 0.049). *Latescibacterota* was negatively correlated with Actinobacteriota (SC: -0.86 ; p -value: 0.001) and having positive correlation with Chloroflexi (SC: 0.69; p -value: 0.025), Nitrospirota (SC: 0.82; p -value: 0.003) and *Desulfobacterota* (SC: 0.66; p -value: 0.037). Spirochaetota is positively correlated with Chloroflexi (SC: 0.85; p -value: 0.001) and *Desulfobacterota* (SC: 0.91; p -value: 0.002) (Fig. 4A; Supplementary File 3).

At the genus level, *Pseudomonas* showed significant positive correlation with *Gemmatimonas*, *Candidatus division_A0839*, *Candidatus division_SM1A02*, *Latescibacterota* (SC: > 0.8 , p -value < 0.05). *Gemmatimonas* showed significant positive correlation with *Candidatus division_A0839*, *Candidatus division_SM1A02* and negatively correlated with *Candidatus division_WCHB1-41* (SC: > -0.7 , p -value < 0.05). *Candidatus division_A0839* was positively correlated with *Pseudomonas*, *Gemmatimonas*, *Candidatus division_SM1A02* (SC: > 0.7 , p -value < 0.05) and negatively correlated with *Latescibacterota*, *Candidatus division_BSV6* (SC: > -0.7 , p -value < 0.05). *Latescibacterota* showed positive correlation with *Candidatus division_MND1*, *Paludibaculum*, *Candidatus division_SBR13*, *Luteitalea* (SC: > 0.6 , p -value < 0.05), and negative correlation with *pseudomonas*, *Candidatus division_A0839* (SC: > -0.6 , p -value < 0.05). Remaining profile is presented in Supplementary File 3 and in Fig. 4B.

Comparative analysis between water and sediment samples

Antibiotic resistance pathways analysis

Comparison of antibiotic resistance pathways between sediment and water samples revealed that sediment sample

Fig. 2 Unique (unshared) phyla and genera (> 0.1% abundance) present in the water and sediment samples of the Ganga River



harbor relatively high antibiotic resistance pathways in the prevailing bacterial communities. Of the ten antibiotic resistance pathways, seven pathways [β -lactam (50.50%), vancomycin resistance (29.21%), aminoglycosides resistance (0.49%), quinolone resistance (0.15%), phenicol resistance (0.05%), sulphonamide (0.07%), and trimethoprim resistance (0.0002%)] showed relatively higher abundance among the sediment samples. Whereas, remaining three pathways [multidrug resistance (21.68%), tetracycline (1.12%) and macrolide resistance (0.36%)] were prevalent in the water samples (Fig. 5A; Supplementary File 4).

Further analysis at the ARGs level revealed that 76 different ARGs were shared between water and sediment samples

of the 175 total ARGs (Supplementary File 5). Of these 76 ARGs 32 ARGs were significantly different between sediment and water samples using *T*-test (Supplementary File 6; Table 2; Fig. 5B).

Comparative analysis revealed that significantly different vancomycin resistance ARGs were exclusively present in sediment samples, whereas aminoglycosidase resistance ARGs were present in the water samples only. Multidrug resistance was highly prevalent in water sample with only one gene (*marA*; *AraC* family transcriptional regulator, *mar-sox-rob* regulon activator) that showed its dominance in sediment samples. β -lactam ARGs were equally

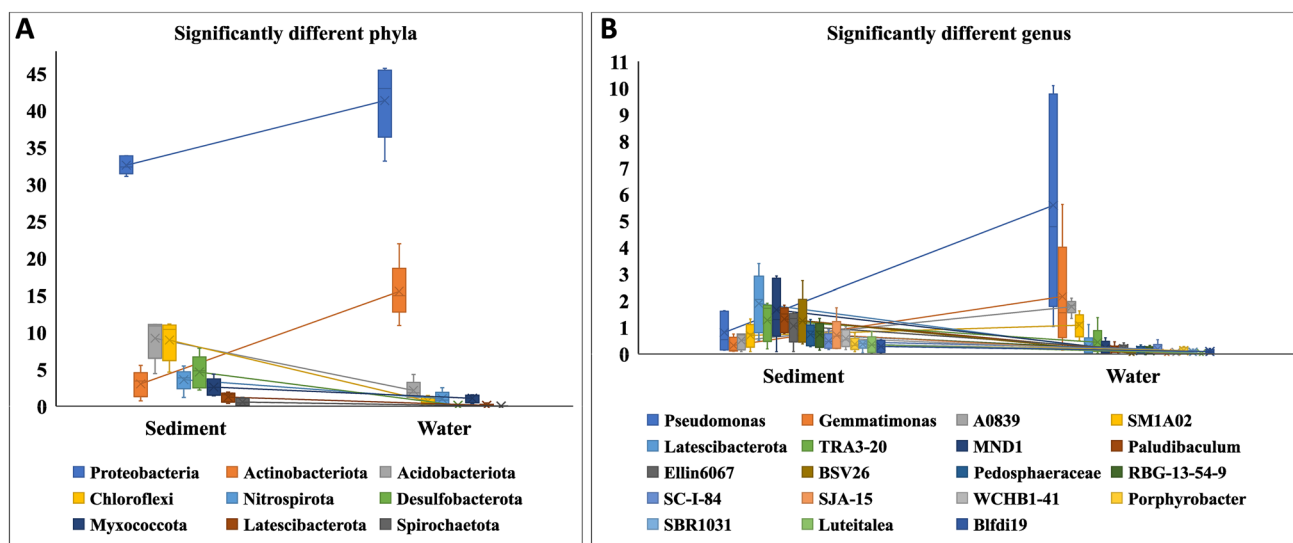


Fig. 3 T-test statistical analysis showed nine and nineteen significantly different (p -value < 0.05) phyla (A) and genera (B) respectively in the water and sediment samples of the Ganga River

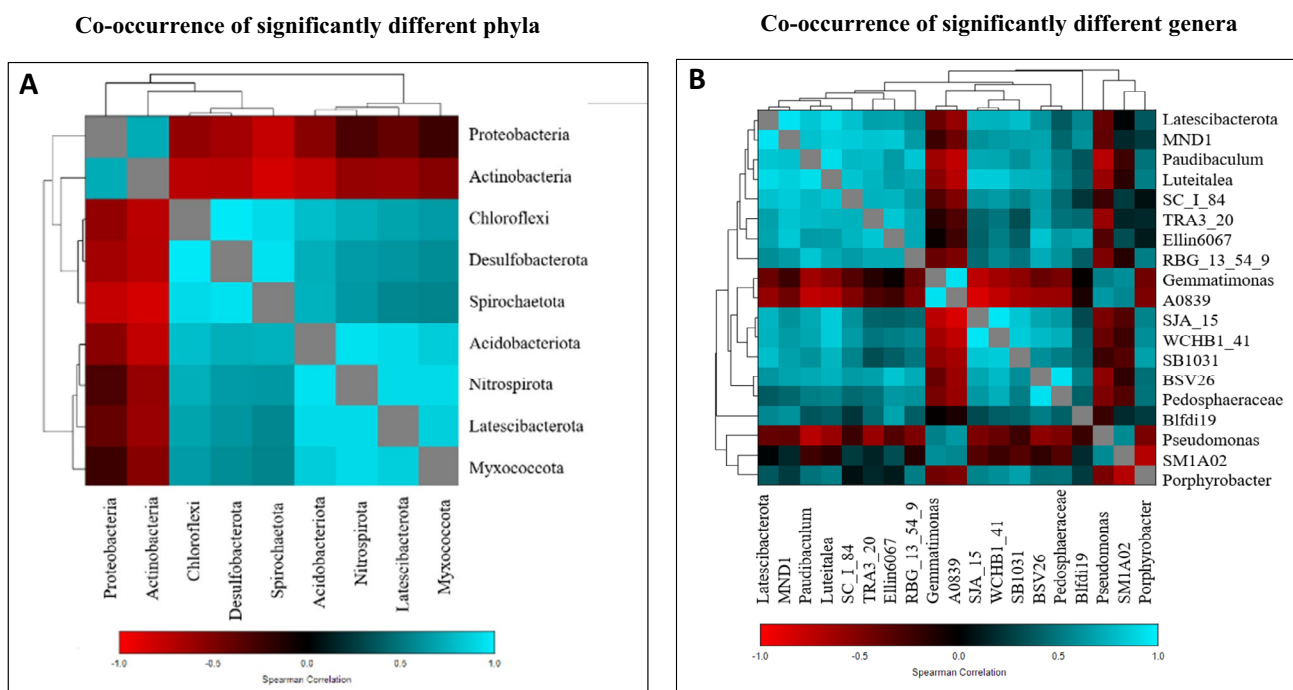


Fig. 4 Co-occurrence and correlation existing between the significantly different phyla (A) and genera (B) observed in the water and sediment samples of the river Ganga

distributed in water and sediment samples both whereas genes conferring resistance to both beta-lactam and multi-drug resistance was more dominant in water samples with the prevalence of only two genes (*tolC*; outer membrane protein; *ompF*; outer membrane pore protein F) in the sediment samples (Table 2).

Principal coordinate analysis

The PcoA analysis using Bray–curtis dissimilarity index revealed high dissimilarity of ARGs between water and sediment samples. More clustering was observed among sediment samples with an overlap of two water samples (MW and RW) in the context of ARGs dissemination. It indicates

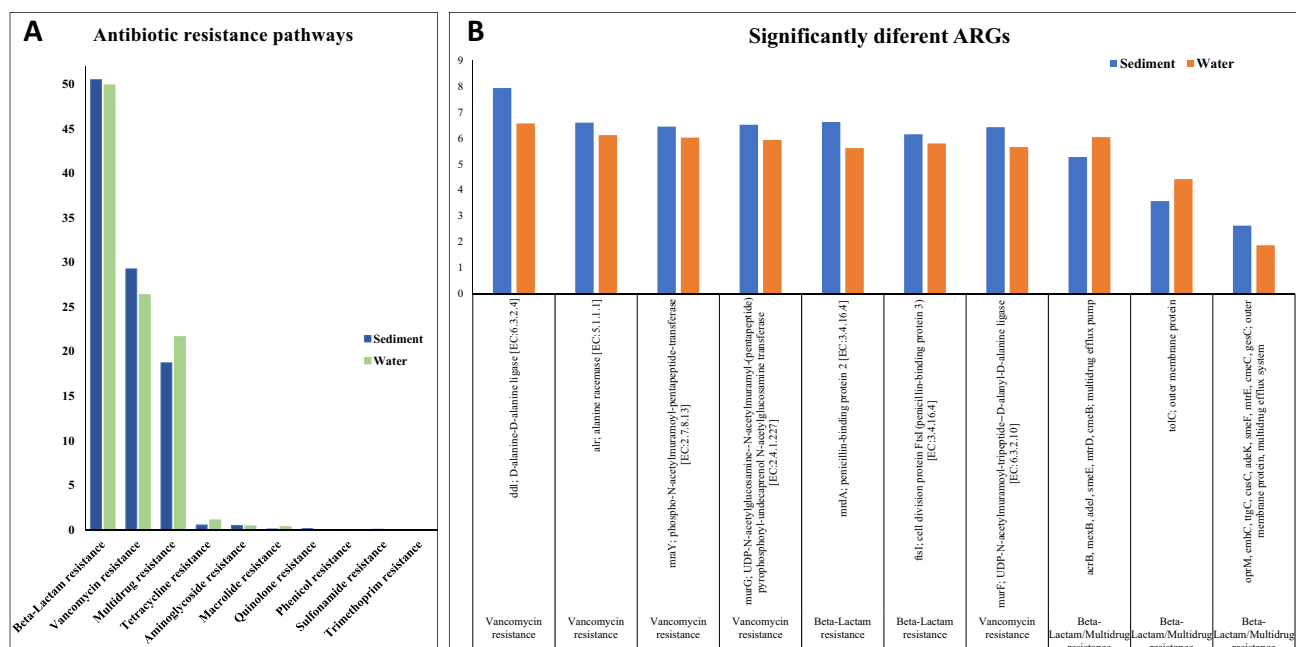


Fig. 5 Various antibiotic resistance pathways observed in the water and sediment samples of the Ganga River (A). Top ten ARGs that were significantly different between water and sediment samples of the Ganga River by using *T*-test statistical analysis (B)

that Mirzapur and Rishikesh water samples showed similar kind of ARGs abundance with sediment samples as compared to other sites of the water samples. However, PCoA analysis at the species level between water and sediment samples indicates that water and sediment samples consist highly dissimilar species diversity (Fig. 6A, B).

Correlation and network analysis between shared genera and antibiotic resistance genes (ARGs)

Of the total 14 genera that showed their relative abundance > 1% among 53 shared genera, five genera (*Flavobacterium*, *Pseudomonas*, *Acinetobacter*, *Candidatus_division CL5003*, and *Candidatus_division SWB02*) showed significantly positive correlation with six antibiotic resistance pathways (β -lactam, vancomycin, multidrug resistance, tetracycline, aminoglycoside, and macrolide resistance pathways) followed by *Cyanobium_PCC-6307*, *Brevundimonas*, *Gemmatimonas*, *Candidatus_division A0839*, and *Gemmatimonas* showing significantly positive correlation with five different antibiotic resistance pathways (β -lactam, vancomycin, multidrug resistance, tetracycline, and macrolide resistance pathways). *Candidatus_division OM190* was observed to be positively correlated with four different pathways (β -lactam, vancomycin, multidrug resistance, tetracycline) whereas three genera *Rheinheimera*, *Rhodobacter* (β -lactam, vancomycin, and multidrug resistance) *Kapabacteriales* (tetracycline, aminoglycoside and macrolide resistance) showed positive correlation with three antibiotic resistance

pathways. Remaining correlation profile of bacterial genera towards antibiotic resistance pathways is presented in the Table 3 and Supplementary File 7. Overall, the highest positive correlation was observed against multidrug resistance (with 13 genera) pathways followed by β -lactam and vancomycin (with 12 genera), tetracycline and macrolide (with 10 genera), aminoglycosides (with 5 genera) and least was against trimethoprim (with only 1 genera) whereas quinolone (with 5 genera) and sulphonamide (with 6 genera) only showed negative correlation amongst the genera showing > 1% abundance (Supplementary File 7). Network analysis between top 14 genera (> 1% abundance) with antibiotic resistance genes (> 1% abundance) showed that five genera (*Nitrospira*, *Hydrogenophaga*, *Pirellula*, *Vicinamibacteraceae* and *Candidatus Kaiserbacteria*) does not show any significant correlation with the ARGs while the remaining nine genera showed significantly high correlation with all the ARGs showing > 1% abundance. The profile of all the 53 shared genera with ARGs are shown in Supplementary File 8. Network consist of 36 nodes (Genus and ARGs) with 399 edges (connection) with a clustering coefficient of 0.425 (Supplementary Fig. S1).

Discussion

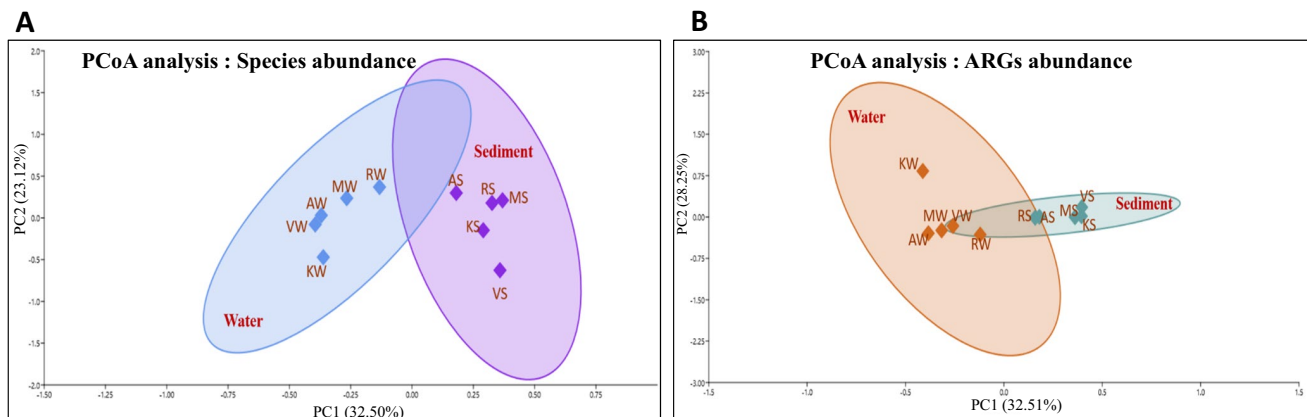
To date, only one report of Reddy and Dubey (2019) discusses about the comparative analysis of water and sediment of the Ganga River bacterial and ARGs profile through

Table 2 Significantly different ARGs present in water and sediment samples of the Ganga River

Antibiotic resistance pathways	Genes and enzymes	Sediment (abundance %)	Water (abundance %)	Relatively dominant
Vancomycin resistance	ddl; D-alanine-D-alanine ligase [EC:6.3.2.4]	7.932323	6.567514	Sediment
Vancomycin resistance	alr; alanine racemase [EC:5.1.1.1]	6.599417	6.117971	Sediment
Vancomycin resistance	mraY; phospho- <i>N</i> -acetylmuramoyl-pentapeptide-transferase [EC:2.7.8.13]	6.447846	6.019872	Sediment
Vancomycin resistance	murG; UDP- <i>N</i> -acetylglucosamine- <i>N</i> -acetyl-muramyl-(pentapeptide) pyrophosphoryl-undecaprenol <i>N</i> -acetylglucosamine transferase [EC:2.4.1.227]	6.516776	5.932606	Sediment
Beta-Lactam resistance	mrdA; penicillin-binding protein 2 [EC:3.4.16.4]	6.624345	5.61703	Sediment
Beta-Lactam resistance	ftsI; cell division protein FtsI (penicillin-binding protein 3) [EC:3.4.16.4]	6.146996	5.795435	Sediment
Vancomycin resistance	murF; UDP- <i>N</i> -acetylmuramoyl-tripeptide-D-alanyl-D-alanine ligase [EC:6.3.2.10]	6.424685	5.65705	Sediment
Beta-lactam/multidrug resistance	acrB, mexB, adeJ, smeE, mtrD, cmeB; multidrug efflux pump	5.272437	6.04154	Water
Beta-lactam/multidrug resistance	tolC; outer membrane protein	3.574139	4.418275	Water
Beta-lactam/multidrug resistance	oprM, emhC, ttgC, cusC, adeK, smeF, mtrE, cmeC, gesC; outer membrane protein, multidrug efflux system	2.627871	1.878969	Sediment
Multidrug resistance	mexF; multidrug efflux pump	0.310701	0.673781	Water
Multidrug resistance	mexE; membrane fusion protein, multidrug efflux system	0.135667	0.339902	Water
Multidrug resistance	oprN; outer membrane protein, multidrug efflux system	0.077821	0.239164	Water
Beta-lactam resistance	ompU; outer membrane protein OmpU	0.074197	0.234824	Water
Multidrug resistance	cueR; MerR family transcriptional regulator, copper efflux regulator	0.05642	0.23173	Water
Aminoglycoside resistance	strB; streptomycin 6-kinase [EC:2.7.1.72]	0.052355	0.146983	Water
Aminoglycoside resistance	aacA; aminoglycoside 6'- <i>N</i> -acetyltransferase [EC:2.3.1.82]	0.038241	0.104616	Water
Aminoglycoside resistance	aac6-I, aacA7; aminoglycoside 6'- <i>N</i> -acetyltransferase I [EC:2.3.1.82]	0.038098	0.087486	Water
Multidrug resistance	evgA, bvgA; two-component system, NarL family, response regulator EvgA	0.029049	0.079177	Water
Beta-lactam resistance	adeR; two-component system, OmpR family, response regulator AdeR	0.020909	0.066682	Water
Beta-lactam/multidrug resistance	adeS; two-component system, OmpR family, sensor histidine kinase AdeS [EC:2.7.13.3]	0.016655	0.060358	Water
Beta-lactam/multidrug resistance	mexX, amrA; membrane fusion protein, multidrug efflux system	0.004071	0.032716	Water
Beta-lactam/multidrug resistance	mexY, amrB; multidrug efflux pump	0.004071	0.032716	Water
Beta-lactam/multidrug resistance	mexZ; TetR/AcrR family transcriptional regulator, mexXY operon repressor	0.004071	0.032368	Water
Multidrug resistance	lasR; LuxR family transcriptional regulator, quorum-sensing system regulator LasR	0.01373	0.020141	Water
Beta-lactam resistance	parS; two-component system, OmpR family, sensor kinase ParS [EC:2.7.13.3]	0.003317	0.022838	Water
Beta-lactam resistance	parR; two-component system, OmpR family, response regulator ParR	0.000812	0.022838	Water
Beta-lactam/multidrug resistance	ompF; outer membrane pore protein F	0.002335	3.17E-05	Sediment
Beta-lactam resistance	ompC; outer membrane pore protein C	0.000665	3.17E-05	Sediment
Multidrug resistance	marA; AraC family transcriptional regulator, mar-sox-rob regulon activator	0.000655	3.17E-05	Sediment

Table 2 (continued)

Antibiotic resistance pathways	Genes and enzymes	Sediment (abundance %)	Water (abundance %)	Relatively dominant
Multidrug resistance	acrF; multidrug efflux pump	0.000655	3.17E-05	Sediment
Beta-lactam resistance	adeC; outer membrane protein, multidrug efflux system	0.012025	0.036267	Water

**Fig. 6** PCoA (Principal coordinate analysis) using Bray–Curtis dissimilarity index to measure the abundance (A) and dissimilarity with respect to bacterial community (A) and ARGs (B) among water and sediment samples of the river Ganga

metagenomic approaches. However, this study was restricted to Varanasi region only (Table 4). Zhang and co-workers also investigated the ARGs profile of the Ganga River water, however, it does not compare the sediment's ARGs profile of the river. Both studies though dig out some information at ARG level, however, portray the partial information of the Ganga River bacterial communities and their antibiotic resistance potential. The main objective of this investigation was to compare the shared and unique bacterial taxa as well as ARGs profile between the sediment and water of the Ganga River to understand the dynamics of bacterial communities and ARGs flow. Besides, the investigation covers a longer stretch of the Ganga River that accounts ~70% region of its total length. While comparing the microbiome at phyla level, six phyla (Proteobacteria, Bacteroidota, Actinobacteriota, Acidobacteriota, Gemmatimonadota, Nitrospirota, and Firmicutes) were shared between water and sediment. It corroborates the phyla level findings on Ganga River water (Reddy and Dubey 2019; Zhang et al. 2019) and sediments (Kumar et al. 2021; Srivastava and Verma 2023a). Similarly, a higher count of common genera (691) was observed between water and sediment, which hypothesize that water and sediment share a high level of bacterial diversity of common taxa both at phyla and genera type. It was also observed that sediment samples exhibited higher bacterial diversity by exhibiting 247 unique genera over the water samples that harboured only 143 unique genera. However, their overall

abundance was below 1% and thus their contribution of these genera at the physiology level might be non-significant and need to be further explored.

Of the nine significantly different phyla, only two phyla (Proteobacteria and Actinobacteria) were more prevalent with the water samples. Zhang et al. (2019) also reported the dominance of these phyla with the Ganga River water. Proteobacteria and Actinobacteria constitute the core bacterial microbiome of the fresh water bodies (Lee et al. 2016). Proteobacteria belong to one of the diverse phyla exhibiting autotrophy coupled to oxidation of sulfur, methane, and hydrogen along with sulfate reduction, and denitrification (Zhou et al. 2020; Gupta et al. 2017). Similarly, Actinobacteria exhibit ecological significance in degrading complex polymers, recycling compounds, and producing bioactive molecules (Mawang et al. 2021; Qin et al. 2019). The prevalence of Acidobacteriota, Chloroflexi, Nitrospirota, and Desulfobacterota corroborate the findings of Huang et al. (2019) where higher count of these phyla was observed. Huang et al. (2019) correlate the abundance of these phyla with high phosphorous and nitrogen content in the sediment samples over the water. *Latescibacterota* and *Spirochaetota* were also comparatively higher in Ganga River sediment, however less information is available for their presence in the sediment of the Ganga River and it need to be explored.

Of the nineteen significantly different genera, only two (*Pseudomonas* and *Gemmatimonas*) were dominant in water

Table 3 Spearman correlation between top 20 abundant genera and antibiotic resistance pathways

S. No.	Genus	Antibiotic pathways (significantly positive correlation, p -value < 0.05)	Antibiotic pathways (significantly negative correlation, p -value < 0.05)
1.	<i>Flavobacterium</i>	β -lactam, vancomycin, multidrug resistance, tetracycline, aminoglycoside, and macrolide resistance pathways	NIL
2.	<i>Pseudomonas</i>	β -lactam, vancomycin, multidrug resistance, tetracycline, aminoglycoside, and macrolide resistance pathways	NIL
3.	<i>Nitrospira</i>	NIL	Trimethoprim resistance
4.	<i>Rheinheimera</i>	β -lactam, vancomycin, multidrug resistance	NIL
5.	<i>Kapabacteriales</i>	Tetracycline, aminoglycoside, and macrolide resistance pathways	Quinolone resistance Sulfonamide resistance
6.	<i>Acinetobacter</i>	β -lactam, vancomycin, multidrug resistance, tetracycline, aminoglycoside, and macrolide resistance pathways	NIL
7.	<i>Cyanobium_PCC-6307</i>	β -lactam, vancomycin, multidrug resistance, tetracycline, and macrolide resistance pathways	Sulfonamide resistance
8.	<i>Candidatus_division OM190</i>	β -lactam, vancomycin, multidrug resistance, and tetracycline	NIL
9.	<i>Brevundimonas</i>	β -lactam, vancomycin, multidrug resistance, tetracycline, and macrolide resistance pathways	Quinolone resistance, Sulfonamide resistance
10.	<i>Hydrogenophaga</i>	Trimethoprim Resistance	NIL
11.	<i>Gemmatimonas</i>	β -lactam, vancomycin, multidrug resistance, tetracycline, and macrolide resistance pathways	Quinolone resistance
12.	<i>Candidatus_division A0839</i>	β -lactam, vancomycin, multidrug resistance, tetracycline, and macrolide resistance pathways	Quinolone resistance, Sulfonamide resistance
13.	<i>Candidatus_division OLB12</i>	NIL	NIL
14.	<i>Pirellula</i>	NIL	NIL
15.	<i>Candidatus_division CL500-3</i>	β -lactam, vancomycin, multidrug resistance, tetracycline, aminoglycoside, and macrolide resistance pathways	Sulfonamide resistance
16.	<i>Vicinamibacteraceae</i>	NIL	NIL
17.	<i>Candidatus_Kaiserbacteria</i>	Aminoglycoside resistance	NIL
18.	<i>Rhodobacter</i>	β -lactam, vancomycin, and multidrug resistance	NIL
19.	<i>Candidatus_division SWB02</i>	β -lactam, vancomycin, multidrug resistance, tetracycline, aminoglycoside, and macrolide resistance pathways	Quinolone resistance, Sulfonamide resistance
20.	<i>Luteolibacter</i>	NIL	NIL

samples. The prevalence of *Pseudomonas* spp. in Ganga water is not surprising and has been consistently reported from various fresh water bodies (Batrach et al. 2019; Khan et al. 2007; Drucker and Panasyuk 2006; Morris et al. 2008; Pietsch et al. 2017). *Pseudomonas* prevalence in the Ganga River may be due to the mass human bathing during the several religious practices in India throughout the year. *Pseudomonas* spp. are considered as a normal bacterial flora of humans with a colonization rate of ~2% in skin, ~3.3% in nasal mucosa, 0–6% in the throat and 2–24% in faecal sample which might increase to 50% during hospitalization (Morrison and Wenzel 1984). *Pseudomonas* showing relatively higher dominance in water indicates the contamination of the hospital as well as human waste in the river water samples of the Ganga River. Similarly, *Gemmatimonas* spp. are reported to possess natural resistant to antibiotic such as ampicillin, bacitracin, chloramphenicol (Zeng et al. 2015, 2021; DeBruyn et al. 2013) which has relatively higher

abundance in the water samples of the Ganga River may be the consequence of over mineralization. To date, only five species of *Gemmatimonas* have been reported where their physiology is less understood (Zhang et al. 2003). Prevalence of these antibiotic resistant bacteria in water samples is a serious concern as many people consider Ganga River water as pure and used it for domestic chores. Genus *Latescibacterota* showed relatively higher abundance in the sediment sample is also reported to act as an ecological indicator which plays role in biodegradation of xenobiotic compounds (Srivastava and Verma 2023a). *Paludibaculum* was also more dominant in sediment over the water. This facultative anaerobe exhibits capabilities to reduce nitrogen and iron (Hedrich et al. 2011) and indicates for the higher abundance of nitrogen in the Ganga River sediment. The remaining significantly different phyla showed their prevalence below 1%.

ARGs level analysis showed seventy-six antibiotic resistance genes shared among water and sediment samples, of

Table 4 Comparison of various metagenomic studies used to investigate bacterial diversity and antibiotic resistance in the river water and sediment samples of the Ganga and Yamuna River

Sample type	Region	No. of samples taken	Sequencing approach/platform and pipeline	Reported antibiotic resistance pathways	Abundant ARGs in water samples	Abundant ARGs in sediment samples	Abundant phyla	Abundant genera	Conclusions	References
Water and sediment	Varanasi, Uttar Pradesh (5 different ghats of Varanasi)	5 water and 5 sediment samples (comparative study)	Shotgun metagenomic approach and Illumina NextSeq 500 sequencer (MG-RAST) v.4.03 server	beta-lactam, multidrug/efflux, Elfa-mycin, aminoglycoside, tetracycline, amino-coumarin, mupirocin, polymyxin, tetracycline, chloramphenicol, and rifampicin	High abundance of ARGs subtypes namely <i>ileS</i> , <i>mfd</i> , <i>parY</i> , <i>mtrA</i> , and <i>mdt</i> was detected in majority of the water samples	High abundance of <i>acrB</i> , <i>acrF</i> , <i>acrD</i> , <i>mexB</i> , <i>AdeI</i> , <i>sulI</i> , <i>sneE</i> and <i>msbA</i> was detected in majority of the sediment samples	Proteobacteria, Actinobacteria, Firmicutes, Bacteroidetes and Chloroflexi	<i>Arcobacter</i> , <i>Methanosaeta</i> , <i>Lepidotrichia</i> , and <i>Pseudomonas</i> to be the most dominant taxa in sediment, <i>Terribacterium</i> , <i>Acidibacter</i> , <i>Acinetobacter</i> were the most dominant taxa in water	ARGs was found to be high in sediment than water samples; most of the ARGs significantly correlated with <i>Arcobacter</i> , <i>Bacteroides</i> , <i>Methanobacterium</i> , <i>Pseudomonas</i> , <i>Parabacteroides</i> and <i>Acidibacter</i>	Reddy and Dubey (2019)
Water	Yamuna River, New Delhi, India	2 water samples of two different season wet and dry	Illumina MiSeq (Amplicon sequencing). QIIME PIPELINE	Rifampin, macrolide, chloramphenicol, tetracycline, phenicol, amino-coumarin, β -lactams, lipopeptides, elbamycin, polymyxins, aminoglycosides, isoniazid, trimethoprim, lincosamide	Presence of genes for penicillin binding, peptidoglycan biosynthesis inhibition, β -lactamase via <i>AmpG</i> , <i>AmpR-AmpC</i> and <i>Opp-Blal</i> , <i>Blaz</i> , RND efflux pumps for efflux of the β -lactams	-NA-	Proteobacteria, Bacteroides and Actinobacteria	<i>Acinetobacter</i> , <i>Aeromonas</i> , <i>Flavobacterium</i> , <i>Acinetobacter</i>	River water possesses many antibiotic resistance genes. mainly belong to efflux systems, showing resistance to multiple antibiotics. Metallo- β -lactamases and <i>NDM</i> genes was found in abundance	Mittal et al. (2019)

Table 4 (continued)

Sample type	Region	No. of samples taken	Sequencing approach/platform and pipeline	Reported antibiotic resistance pathways	Abundant ARGs in water samples	Abundant ARGs in sediment samples	Abundant phyla	Abundant genera	Conclusions	References
Sediment	Yamuna River, New Delhi	3 samples sites	Shotgun metagenomic approach and Illumina TruSeq Nano (Res's finder server)	Aminoglycoside, beta-lactam, colistin, fluoroquinolone, fosfomycin, fusidic-acid, glycopeptide, MLS, nitroimidazole, oxazolidinone, phenicol, rifampicin, sulphamamide, tetracycline, and trimethoprim antibiotic	BLR genes in Group-1 (<i>blaOXA-10</i> , <i>blaOXA-129</i> , and <i>blaOXA-347</i>), MR genes in Group-2 (<i>erm</i> (G) and <i>erm</i> (A)), <i>sul2</i> , TET genes (<i>tetA</i> and <i>tetG</i>), and PR genes (<i>floR</i> , and <i>cmlA1</i>)	-NA-	-NA-	Genus not available. Abundance of species such as <i>Pseudomonas aeruginosa</i> , <i>Shigella sonnei</i> , <i>Escherichia coli</i> , <i>Enterococcus faecium</i> , <i>Vibrio cholerae</i> , <i>Streptococcus pneumoniae</i> , was observed	11 different ARGs were identified in the sediment metagenome of the Yamuna River, of which the <i>AGR</i> , <i>BLR</i> , <i>MR</i> , and <i>TET</i> genes were found to be highly abundant	Das et al. (2020)

Table 4 (continued)

Sample type	Region	No. of samples taken	Sequencing approach/ platform and pipeline	Reported antibiotic resistance pathways	Abundant ARGs in water samples	Abundant ARGs in sediment samples	Abundant phyla	Abundant genera	Conclusions	References
Water and sediment	Uttarakhand and Uttar Pradesh region (Ganga River)	10 sites of 6 cities (20 pooled samples) (Comparative study)	Illumina MiSeq (Amplicon sequencing). QIIME2 PIPELINE	β -lactam, vancomycin, aminoglycosides, quinolone, phenicol, sulphonamide, trimethoprim, multidrug tetracycline macrolide resistance was observed	Aminoglycosidase and Multidrug resistance ARGs was highly prevalent in water samples only	Vancomycin resistance ARGs were exclusively present in sediment samples, <i>marA</i> ; <i>AraC</i> family transcriptional regulator, mar-sox-regulon activator, <i>tolC</i> ; outer membrane protein; <i>ompF</i> ; outer membrane pore protein F) was observed in sediment samples only	<i>Proteobacteria</i> , <i>Actinobacteria</i> , <i>Bacteroidetes</i>	<i>Flavobacterium</i> , <i>Pseudomonas</i> , <i>Acinetobacter</i>	Sediment harbours higher level of species richness and ARGs over the water samples. The members of <i>Flavobacterium</i> , <i>Pseudomonas</i> , <i>Acinetobacter</i> , <i>Candidatus_division CL5003</i> , and <i>Candidatus_division SWB02</i> may be considered as reservoir for such ARGs	Present study

NA not available

which twenty-two ARGs showed > 1% abundance. Where, β -lactam, vancomycin, and multidrug resistance genes, tetracycline and aminoglycoside constitute the top five prevalent antibiotic resistance pathways in between water and sediment. Prevalence of β -lactam resistance genes have already been reported from various rivers of India (Mirzaei et al. 2018; Li et al. 2018; Mutiyar and Mittal 2014; Ahammad et al. 2014). Detection of *bla*_{NDM-1} genes from the surface water of Yamuna River (a tributary river of the Ganga which joins in Allahabad) was first time reported from India in 2010 (Walsh et al. 2011). Thereafter, its dissemination was identified in the upper Gangetic regions of Rishikesh and Haridwar (Ahammad et al. 2014). Heavy pilgrimage visits during seasons of May–July were assumed as a reason for dissemination of *bla*_{NDM-1} genes in Ganga River (Ahammad et al. 2014). *bla*-TEM genes were also reported in water samples of Brahmaputra River of India by using culturable approach (Kumar et al. 2020). The Brahmaputra River and their associated water bodies have also been analysed for the prevalence of ciprofloxacin and tetracycline resistant *Vibrio cholerae* during the flood (Bhuyan et al. 2016). Second dominant load was of vancomycin resistance genes in the present investigation that were comparatively higher in water samples. Lata et al. (2009) reported the elevated level of *Enterococci* with emerging vancomycin resistance in surface water bodies. In a recent investigation, Ali et al. (2021) also reported higher abundance of vancomycin due the faecal based pollution in the Ganga River. Horizontal gene transfer (HGT) event was reported for the prevalence of multidrug resistance genes obtained in the isolates of Rishikesh region of the Ganga River (Ali et al. 2021). Besides, the downstream Ganga region was also observed for the presence of *Bacillus subtilis*, *Clostridium difficile*, and *Neisseria* spp., that were absent in upstream region of the Ganga River. Several unique ARGs were also observed between water and sediment, however, their abundance was below 1% among the respective samples and thus find less impact for their role in dynamics of ARGs of the river.

Of the significantly different ARGs (p -value < 0.05), 32 ARGs were significantly different between water and sediment samples. Where, vancomycin resistance genes were more associated with sediment samples that recapitulates our previous findings on sediment of the Ganga River (Srivastava and Verma 2023a). Vancomycin is considered as the last reservoir antibiotic against several bacterial infection, therefore, prevalence of vancomycin resistance genes the Ganga River water is a fright. Various environments such as domestic and wild animal faeces (Harwood et al. 2002; Oravcova et al. 2014; Klibi et al. 2015), waste water treatment plants (Adegoke et al. 2022), sewage water (Nishiyama et al. 2015; Lanthier et al. 2010), hospital discharge (Girijan and Pillai 2021; Novais et al. 2005) and surface water (Ben Said et al. 2015; Nishiyama et al. 2015; Lat et al. 2009)

have been reported as the major contributors for the dissemination of the vancomycin ARGs. It indicates the inappropriate consumption of vancomycin especially in India, where vancomycin is considered as one the most potential drug to treat the *Staphylococcus aureus* infection (NCDC, India report, 2016). In India, Tiwari, and Sen (2006) were the first to report the vancomycin intermediate *Staphylococcus aureus* (VISA) and vancomycin resistant *Staphylococcus aureus* (VRSA) from tertiary care hospitals of northern Indian region.

Water samples were more associated with aminoglycosidase resistance genes. Reddy and Dubey (2019), however, have reported a very low abundance of aminoglycoside resistance genes in the Varanasi region of the Ganga River water. This variation may be due to the selection of only one region of the river. Various genes of β -lactam ARGs were also more prevalent in water over the sediment which corroborates the findings of Reddy and Dubey (2019) on a short stretch of the Varanasi region of the Ganga. However, in contradict to Reddy and Dubey 2019, we observed a relative high abundance of significantly different ARGs in the water sample as compared to the sediment samples. PCoA analysis revealed higher diversity of ARGs among water samples over the sediment and identified them as distinguished spatial clusters. Besides, it also shared clustering with Mirzapur and Rishikesh water samples. The clustering of water with the sediment samples indicates that there might be dissemination of ARGs occurring between sediment and water or vice-versa due to low flow rate of water, seasonal effect, accumulation of extracellular DNA and a shift in microbial community (Calero-Cáceres et al. 2017; Mao et al. 2014). A gradual increase in the abundance of several antibiotic resistance pathways has been observed from the samples of Uttarakhand to Uttar Pradesh region which might be due to high pollution load and anthropogenic activities in the Uttar Pradesh sites. More research is required in the line using traditional approaches of microbiology on inhabitant bacterial isolates for better understanding of HGT events for dissemination of ARGs. Network based analysis in 14 shared genera (> 1%) between water and sediment samples indicate that five genera five genera (*Flavobacterium*, *Pseudomonas*, *Acinetobacter*, *Candidatus_divison* CL5003, and *Candidatus_division* SWB02) may be considered as the reservoir of ARGs. It chiefly includes genes of six antibiotic resistance pathways (β -lactam, vancomycin, multidrug resistance, tetracycline, aminoglycoside, and macrolide resistance pathways). Therefore, Ganga River water samples have profusion of ARGs consist of beta-lactam, multidrug and tetracycline antibiotics. Of them, *Flavobacterium*, *Pseudomonas*, and *Acinetobacter* have already been reported as known opportunistic pathogens (Manfredi et al. 1999; Moradali et al. 2017; Howard

et al. 2012; Antunes et al. 2014) along with their resistance to several antibiotics (Dehbashi et al. 2018; Smiline et al. 2018; Izzet et al. 2019; Wang et al. 2020; Declercq et al. 2021; Sarigül and Koç Çelik 2022).

Conclusion

Overall, water and sediment of the Ganga River unveiled several shared and unique taxa at the level of phyla and genera. It was further observed that the sediment harbours higher level of species richness over the water samples. Genes of antibiotic resistance pathways were also relatively more prevalent among sediment samples over the water. Vancomycin resistance genes were significantly more abundant with the sediment samples, whereas β -lactam resistance genes were equally shared in water and sediment. It indicates that the sediment harbours a reservoir of ARGs by providing a less disturbed niche to bacteria for evolving mechanisms to cope up with present threats of antibiotics. The members of *Flavobacterium*, *Pseudomonas*, *Acinetobacter*, *Candidatus_divison* CL5003, and *Candidatus_division* SWB02 may be considered as reservoir for such ARGs. This investigation, therefore, paves a way to study the ARGs profile among these candidates using traditional cultivation approaches along with their dissemination mechanisms. The findings can be further employed to make the policies for regulating the consumption as well as disposal of reported ARGs here.

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Declarations

Conflict of interest The authors declare no conflict of interest.

Consent for publication On acceptance of the manuscript the copyright from the author will be transferred to the journal.

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